

Group Project

RegTech Architecture for OTC Derivatives: Automated Cross-Border Compliance & Taxonomy Expansion.

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Percentage of Contributions by Each Member

Member Name	Percentage of Contributions	Justifications
Wang Yue	25%	Responsible for all the code and Section 1 Regulatory Landscape of the written report, introducing the global OTC reporting framework, the three identifier pillars and regimes' relationships. Refined and delivered Segment 1 Live Demo of the recorded presentation.
Yu Xuerui	25%	Responsible for Section 2 System Architecture and Section 3 Portfolio Findings of the written report, focusing on integrating technical descriptions and compiling portfolio audit metrics. Refined and delivered Segment 2 One Finding of the recorded presentation.
Peng Yulong	25%	Responsible for section 4 Classification Analysis: Prediction Contracts and the Reporting Frontier of the written report. Wrote and delivered Segment 3 Policy Argument of the recorded presentation.
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Section 1: Regulatory Landscape

The global Over-the-Counter (OTC) derivatives market is large, structurally complex, and features prominent cross-institutional, cross-market and cross-jurisdictional features. To effectively guard against systemic financial risks and enhance overall market transparency, the Financial Stability Board (FSB) and the International Organization of Securities Commissions (IOSCO) have promoted and established a standardized regulatory reporting framework built around three core identifiers: LEI, UTI and UPI. This framework lays a solid and unified foundation for global regulatory data governance and cross-border supervisory cooperation.

The Legal Entity Identifier (LEI) acts as a globally unique identifier for legal entities for institutional participants, uniquely identifying legal entities engaged in financial activities. It guarantees that all trading parties can be fully identified, traced and supervised transparently, becoming a mandatory reporting field under both CFTC and EMIR rules and the basic support for cross-jurisdictional data alignment. The Unique Transaction Identifier (UTI) assigns an exclusive global serial number to every OTC derivatives trade, allowing precise recognition and verification across different institutions, technical systems and regulatory regimes, while effectively avoiding duplicate submission and omission of transaction reports. The Universal Product Identifier (UPI) unifies classification and definition of derivative products through standardized templates, eliminating inconsistent professional terminology among markets and institutions and realizing unified product identification worldwide.

The three identifiers support and coordinate with one another, forming the core data infrastructure of global regulatory reporting. Under this overarching unified framework, leading jurisdictions represented by the US CFTC and EU EMIR have formulated differentiated implementation guidelines and reporting rules based on their own market conditions. The CFTC prioritizes clearing status disclosure and real-time data submission, whereas EMIR sets more rigorous constraints on margin arrangements, collateral allocation and counterparty credit risk assessment.

Such regulatory disparities reflect varied risk control focuses of different authorities, meanwhile creating market asymmetries and clear incentives for regulatory arbitrage. With cross-border OTC transactions growing rapidly in volume and complexity, market participants face mounting pressure to comply with fragmented reporting obligations across regions. Many small and medium-sized financial institutions lack mature technical systems to reconcile divergent regulatory standards, leading to higher operational burdens and potential reporting errors. Building an intelligent automated compliance checking engine compatible with multiple jurisdictional rules is therefore crucial for raising reporting efficiency, cutting compliance operating costs, reducing manual intervention and promoting coordinated global financial regulatory governance.

Section 2: System Architecture

2.1 Global Pipeline Design & Modular Topology

The compliance engine is architected as a modular RegTech ingestion pipeline designed in a decoupled Python framework. The system ingests raw over-the-counter (OTC) transaction records, executes syntactic cleansing, resolves asset classifications, and runs cross-border compliance audits. The pipeline segregates concerns into three independent computational modules linked via structured data objects, isolating data state mutations and guaranteeing that transaction batches flow linearly without pipeline-wide bottlenecks. The structural schema and linear data execution trajectory of this modular configuration are visually synthesized in Figure 1 below.

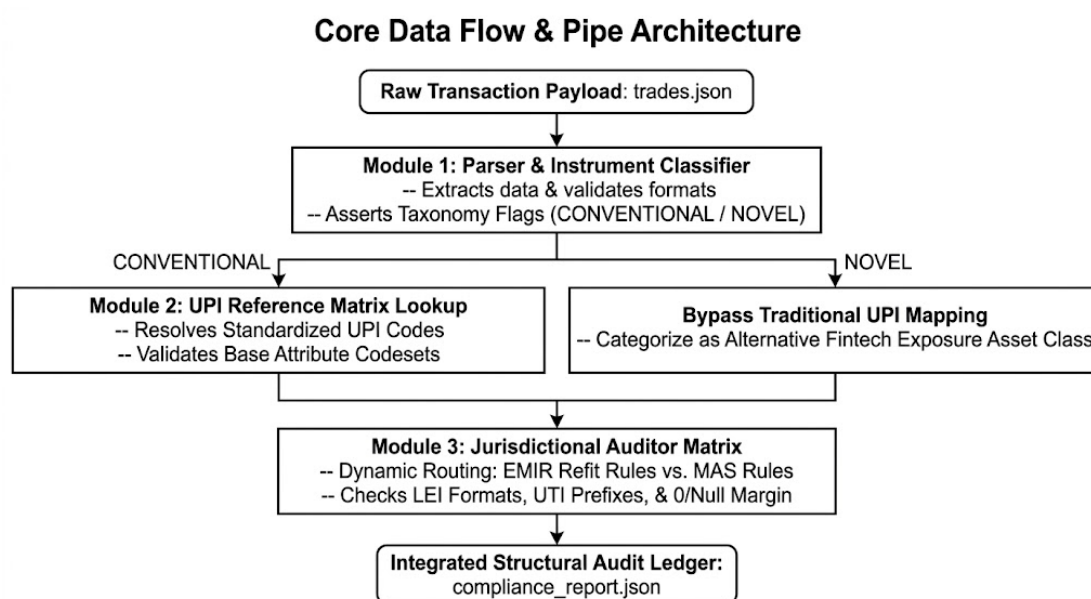


Figure 1

2.2 Module 1: Ingestion, Chronological Parsing, and Taxonomy Classification

Module 1 serves as the primary gateway for data validation. The engine first verifies the absolute presence of four core economic attributes: `trade_id`, `asset_class`, `instrument_type`, and `use_case`. Concurrently, temporal conformity is enforced by submitting the `execution_timestamp` to a strict regular expression compiler:

ISO_UTC_REGEX = re.compile(r"^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z\$")

This pattern enforces strict compliance with the ISO 8601 UTC standard. If a transaction records a timestamp utilizing non-standard whitespace delimiters or omits the standard "T" separator (such as trade T032), the engine isolates the record and flags its parsing status as

PARTIAL to prevent historical data loss while logging a quality warning.

Module 1 then asserts the initial regulatory taxonomy classification flag:

- **CONVENTIONAL:** Assigned to instruments present in standard financial asset classes (Rates, Credit, FX, Equity, Commodities).
- **NOVEL:** Assigned to emergent event or prediction contracts (such as trades T026–T028 and T031) falling outside traditional boundaries.
- **AMBIGUOUS:** Invoked if missing or contradictory fields prevent objective determination.

Chronological safety is maintained via calendar mapping (`datetime.strptime`), incorporating custom routing to bypass open-ended regulatory placeholder codes (such as the "9999-99-99" perpetual notation), preventing date overflow faults during matrix execution.

2.3 Module 2: Reference Data Alignment and UPI Mapping

Following syntactic parsing, conventional transactions enter Module 2 to simulate the International Securities Association's Derivatives Service Bureau (ANNA-DSB) reference library structure. The engine constructs a hierarchical lookup token by compounding primary taxonomy nodes into a compound string: `[AssetClass].[InstrumentType].[UseCase]`.

To maintain high availability and remove remote environment dependencies, the lookup maps directly against an optimized local database cache (`FALLBACK_UPI_DATABASE`). This maps standard transactions to their corresponding ISO 4914 Unique Product Identifier (UPI). If the taxonomy flag is resolved as NOVEL, Module 2 activates a programmatic bypass. Instead of triggering a data data-quality failure for the lack of an ANNA-DSB template, the engine isolates the record and skips traditional attribute codeset checks, cleanly separating operational data errors from structural taxonomy gaps.

2.4 Module 3: Multi-Jurisdictional Compliance Auditor Matrix

Module 3 contains the core validation logic evaluating compliance under overlapping cross-border regimes, specifically European (EMIR Refit) and Singaporean (MAS) reporting guidelines. A dynamic routing layer evaluates transaction variables (such as currency denominations or trading platforms) to stream payloads into their active jurisdictional paths, executing the following specific checks:

1. **LEI Check Digit Audit:** Verifies counterparty identifiers against ISO 17442. For conventional trades, the final two positions of both LEI fields are validated using a

programmatic implementation of the ISO 7064 MOD 97-10 check digit algorithm. Explicit placeholder strings or invalid structures (such as "MISSING_LEI" in trade T004) force an immediate NONCOMPLIANT designation.

2. **UTI Namespace Alignment:** Enforces ISO 23897 pairing rules by performing a character-level slice check. The first 20 characters of the global UTI must match the 20-character LEI of the reporting institution; namespace breaks (such as trade T032) are flagged as critical validation errors.
3. **The Margin Omission Paradox:** Evaluates risk-mitigation data completeness under EMIR and MAS rules. If a contract declares an active collateral portfolio code, the reported initial margin is type-checked. If the active balance is zero, it must be explicitly reported as the integer 0. Leaving the field as a floating null/None (as triggered in T017 and T033) is treated as an illegal omission of mandatory risk metrics, invalidating the trade.
4. **Settlement Currency Boundaries:** Filters currency fields against ISO 4217 asset codes. Alternative asset classes like XAU (gold) are validated as permissible, while decentralized stablecoins like USDC (trades T027 and T033) are blocked as non-compliant reporting formats.

2.5 Fault Tolerance and Graceful Degeneracy

The engine ensures that data failures never trigger pipeline-wide runtime crashes. Module 3 acts as an isolated error aggregation node, utilizing structured try-except blocks that capture and append granular violation vectors directly into an independent nested ledger (findings). Even if a transaction record fails multiple compliance dimensions, the pipeline degrades gracefully, producing a machine-readable output record that ensures complete tracking visibility without interrupting portfolio processing.

Section 3: Portfolio Findings

3.1 Overall Composition & Compliance Overview

The compliance engine executed an end-to-end audit across the entire 33-trade portfolio. Telemetry indicates an overall compliance rate of 84.8% (28 transactions successfully passed), with 5 non-compliant anomalies intercepted (15.2%). In terms of asset distribution, conventional derivatives (Rates, Credit, FX, Equity, Commodities) comprise 87.9% (29 contracts), while novel event contracts represent 12.1% (4 contracts). This quantitative breakdown highlights a dual-track portfolio structure dominated by traditional hedging instruments alongside

alternative fintech risk exposures.

Metric Category	Evaluation Dimension	Trade / Contract Count	Percentage / Proportion
Portfolio Scope	Total Portfolio Trades Audited	33	100.0%
	└─ Conventional Derivatives	29	87.9%
	└─ Novel Event Contracts	4	12.1%
Audit Outcomes	● Regulatory Compliant Pass	28	84.8%
	● Non-Compliant Intercepted	5	15.2%

3.2 Counterparty Data Omission and Structural Identification Defects

Among the intercepted conventional derivatives, equity option trade T004 failed due to a critical identification breach. The counterparty legal entity identifier field (`other_counterparty_lei`) was populated with the non-standard placeholder string "MISSING_LEI". This directly violates mandatory disclosure requirements under both EMIR Refit and MAS frameworks, which strictly mandate valid 20-character ISO 17442 codes for bilateral entities, resulting in an automatic rejection of the transaction.

3.3 The Margin Paradox: Explicit Zero (0) vs. Omitted Omissions (Null)

Credit default swap T017 exposed a significant data quality defect regarding risk-buffer reporting. Although the trade declared a valid collateral portfolio code (PORT-H234), the `initial_margin_posted` field was left completely blank (null). Under derivative reporting rules, an active portfolio with zero exchanged margin must explicitly report the integer 0; leaving the field as null constitutes an invalid regulatory filing. Synthetic test case T033 similarly failed due to this margin omission coupled with an unauthorized stablecoin currency denomination.

3.4 Cross-Border Reporting Restrictions & UTI Namespace Inversion

Synthetic edge cases further revealed structural violations under overlapping jurisdictional rules. FX forward T032 was downgraded to a PARTIAL parsing status due to a non-standard whitespace delimiter in its execution timestamp. Furthermore, it triggered a fatal compliance

failure because its Unique Transaction Identifier (UTI) namespace prefix did not match the reporting institution's LEI, violating global trade repository transaction pairing and deduplication mandates.

3.5 Compliance Feasibility of Novel Products & UPI Taxonomy Blind Spots The granular audit of the four novel event contracts (T026, T027, T028, T031) yielded a profound policy insight: three out of the four novel transactions (covering political outcomes, regulatory predictions, and parametric weather risks) achieved 100% data compliance (COMPLIANT) across all data layout, chronological, and identifier fields. Only T027 was blocked due to its utilization of USDC, a non-ISO 4217 decentralized stablecoin. This empirically proves that prediction market instruments can fully conform to rigorous regulatory reporting standards; their current invisibility within traditional financial streams stems not from data quality issues, but from a fundamental taxonomy gap in the international ANNA-DSB UPI registry.

Section 4: Classification Analysis: Prediction Contracts and the Reporting Frontier

This section applies the economic function test from the assignment, draws on Brandes (2026) to analyze the regulatory gap, proposes a concrete UPI schema, and finally reflects on what a compliance engine built on today's taxonomy cannot see.

4.1 Economic Function Test

Brandes (2026) argues that the prohibition of prediction market contracts in most EU member states was never preceded by an examination of their economic function. To assess whether these instruments perform hedging or price discovery analogous to recognized derivatives, we apply the three-question test to each of the three core event contracts.

T026. The identifiable economic actor is a German renewable energy firm whose revenue depends on state election outcomes because the renewable subsidy regime is materially affected by the vote share of the AfD. The firm has a measurable contractual exposure: its profitability declines if the AfD exceeds 30%. The prediction contract, if freely tradeable, would allow that firm to sell contracts when its own exposure increases, thereby offsetting political risk. Moreover, Kalshi's continuously traded prices provide real-time probability estimates that are more responsive than opinion polls and are backed by real capital at risk. Thus T026 has a hedging actor, enables risk management, and contributes to price discovery.

T027. The economic actor is an EU asset manager holding a USD-denominated bond portfolio that is sensitive to unexpected inflation. The manager faces measurable exposure to a high-inflation outcome. The Polymarket contract, despite being accessed via VPN from a jurisdiction that classifies it as illegal gambling, would allow the manager to hedge that inflation risk. The

price of the contract reflects market-derived probabilities that are updated around each BLS release, offering informational value beyond standard inflation swaps because the contract is binary and event-focused. Hence T027 also meets the test, even though its current legal status in Europe is prohibition.

T028. A fintech firm whose compliance costs hinge on whether credit scoring models are classified as high-risk under the EU AI Act is the natural hedging actor. The contract pays USD 1 if ESMA approves that classification before end-2026, exactly matching the firm's exposure. Trading on a CFTC-regulated DCM gives the firm a legitimate hedging tool. The price discovery function is valuable because regulatory decisions are inherently uncertain and binary. Therefore T028 passes the test as well.

Across all three trades, they serve the same hedging and price-discovery functions as traditional derivatives. The absence of a UPI does not reflect any economic difference; it reflects a historical classification gap.

4.2 A Proposed UPI Schema for Event Contracts

If prediction contracts were brought within the ANNA-DSB framework, a new product definition would be required. Following the template structure of existing UPI definitions, we propose the following schema, illustrated for T028:

```
{
  "AssetClass": "EventContract",
  "InstrumentType": "Binary",
  "UseCase": "RegulatoryDecision",
  "Level": "UPI",
  "Attributes": {
    "EventDescription": {"type": "string", "maxLength": 500},
    "EventDeterminationSource": {"type": "string", "enum": ["OfficialGovernmentAgency",
"ExchangeJudgment", "ThirdPartyOracle"]},
    "PayoutCurrency": {"type": "string", "codeset": "ISOCurrencyCode"},
    "PayoutPerContract": {"type": "number", "minimum": 0, "maximum": 1},
    "SettlementDate": {"type": "string", "format": "date"},
    "DeliveryType": {"type": "string", "enum": ["CASH"]}
  }
}
```

The critical validation constraint is that EventDescription must refer to a verifiable, publicly observable outcome. For T026, the description “AfD vote share $\geq 30\%$ in 2025 Bundestagswahl” would be validated against the official election result published by the Bundeswahlleiter. This preserves the structure of existing UPI templates while introducing event-specific

attributes.

4.3 Jurisdictional Arbitrage and Regulatory Design

The same instrument produces three different compliance outcomes today. A German renewable energy firm can legally hedge election risk on Kalshi (CFTC-regulated) but cannot access that hedge from its own office in Sachsen-Anhalt because German law classifies the contract as gambling. However, the firm can use a VPN to reach Polymarket, an offshore platform with no consumer protection. Who benefits? The arbitrage benefits sophisticated actors with technical ability and legal resources, while retail participants are either excluded or exposed to unregulated platforms. The harmed parties include German consumers who cannot access a legitimate risk-management tool and European regulators who lose visibility into political-risk hedging flows.

Brandes proposes five framework elements. We examine two from a RegTech perspective. Operator neutrality: the platform's revenue must be independent of contract outcomes, derived from transaction fees not from taking positions against users. Implementation requires that the trade repository collect fee structure data and verify statistical independence of operator revenue from event outcomes. Validation logic must compare reported fee schedules with position data. Reporting infrastructure would need a new field: `operator_fee_model`. Position limits and consumer protection: retail participation should be subject to limits. Implementation requires that the reporting system aggregate positions per legal entity per event contract, a feature not currently supported by standard UTI/LEI reporting. New data fields would include `event_identifier` and `aggregated_position`.

Bringing prediction contracts under EMIR would require two concrete changes. First, the ANNA-DSB library must adopt the new product definition proposed above, with a new `EventContract` asset class and enumerated `EventType` values. Second, SDR reporting infrastructure would need to add new fields: `event_description`, `event_source`, `settlement_condition`, and `payout_date`.

4.4 Limits of the Compliance Engine

A compliance engine built on the current taxonomy is blind to systemic risks that do not yet have a UPI. Kalshi alone processed over USD 13 billion in notional volume in March 2026 – a volume comparable to some niche commodity derivatives. None of that activity appears in any trade repository report because event contracts have no UPI. The absence of a UPI is therefore itself a systemic risk indicator: it signals an asset class that operates entirely outside the regulatory perimeter. Regulators should treat the lack of a product definition as an urgent data gap.

If we were advising the CFTC on the technical requirements for the prediction market rule-making (ANPR 91 FR 12516), I would prioritize two changes. First, add `EventContract` as a top-level asset class with a minimal set of attributes (event description, determination source, payout currency) to enable immediate reporting. Second, mandate that every event contract

traded on a DCM be assigned a provisional UPI by the exchange until the official library is updated.

Section 5: Limitations and Future Work

Although the engine provides a useful framework for validating OTC trade data, it remains significantly simpler than a production trade reporting system.

The first limitation is that the system validates trades only as point-in-time records. Real reporting systems must track the entire trade lifecycle, including amendments, novations, compressions, cancellations, and corrections. A future version would therefore require a persistent trade state model linked to each UTI so that historical reporting changes can be monitored over time.

A second limitation is the absence of valuation and collateral reporting. Regulatory frameworks such as EMIR Refit, MAS reporting standards, and the CFTC Rewrite require firms to report mark-to-market valuations, collateral portfolio information, initial margin, and variation margin. While the engine checks for the presence of some collateral fields, it does not connect to pricing systems or collateral management platforms. Future development would therefore require integration with live market and collateral data feeds.

The system also lacks bilateral reconciliation functionality. In real reporting environments, both counterparties must report the same UTI so that trade repositories can pair and match the transaction. The current engine validates only one side of the trade and therefore cannot detect pairing breaks or inconsistencies between counterparties' submissions.

Another limitation is the absence of live SDR connectivity. The system produces a local compliance report and dashboard but does not generate machine-readable regulatory submissions or process acknowledgement messages from trade repositories. A production system would need to support formats such as ISO 20022 and maintain a feedback loop for rejected or corrected submissions.

Finally, the project highlights the limits of rule-based compliance systems when instruments fall outside existing taxonomies. Prediction contracts can be identified and flagged, but the engine cannot create a regulatory classification where none exists. Future work should therefore focus on extending product definition libraries to include event-based contracts and developing clearer reporting standards for these products.